

LEVERAGED TRADING VIA LENDING PLATFORMS

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ABSTRACT. Decentralised lending protocols enable users to enter a leveraged long or short trading position, which provides one of the economic rationales for over-collateralised loans. The smaller the haircut on the provided collateral (initial Loan-to-Value), the higher the leverage. This work compares loan positions on leading platforms with perpetual futures, a primary mechanism for trading leverage in a decentralised finance (DeFi) ecosystem. We introduce the notion of the *implied funding fee/funding rate* and contrast it with the funding fee/funding rate for perpetual futures and find that the former is significantly less volatile than the latter. We also show that liquidations on decentralised lending platforms correspond to margin closeouts with the appropriately chosen maintenance margin rule, which we derive. Furthermore, we study PnL for both positions, the likelihood of liquidation for loan positions, and the margin calls for perpetual futures across multiple market conditions. The analysis of historical data and extensive simulations underpins our study.

1. INTRODUCTION

Decentralized loan protocols such as Aave [1], Compound [5], or Morpho [7] allow users to lend and borrow assets on the Ethereum chain without having to rely on a central trusted entity such as traditional banks [15, 14, 13]. These protocols do not rely on credit ratings to assess the probability of borrowers defaulting but instead require that the borrower post collateral in excess on the issued loan. Yield-seeking users deposit their assets into the protocol to earn interest. There are several economic rationales driving demand for over-collateralized loans. In periods of thin liquidity, the execution cost of swapping assets at the exchange may exceed the price of taking a loan with the intention of never repaying it. Furthermore, agents wanting to remain in control of the majority of the token stake in a particular project are able to create liquidity using lending protocols. Crucially, loan protocols enable users to enter a leveraged long and short position. Indeed, by depositing risky assets as collateral and borrowing a stablecoin, an agent using a lending platform has unlimited upside potential but controlled downside risk. If the price of the risky asset goes up, she keeps all the upside by repaying the loan plus interest. If the price of the risky asset goes down, she may simply forfeit the asset by walking away from the loan. The short position is achieved by depositing stablecoins and borrowing risky assets. As such, loan positions provide an alternative to perpetual futures that are used to enter leveraged trading positions and are, by far, the most popular derivative traded in

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Acknowledgements: We would like to thank Samuel N. Cohen (University of Oxford), and David Siska (University of Edinburgh & Vega Protocol) for their feedback on this work, as well as Alice Ng for her research assistance.

cryptocurrency markets [10, 2]. Perpetual futures contracts are designed to track spot prices. To achieve that, long position holders periodically pay short position holders funding fees proportional to the gap between perpetual futures and spot prices. The funding fee incentivizes trades to narrow the futures-spot gap. To better see how loan positions relate to perpetual futures, in this article, we introduce the notion of the *implied funding fee/funding rate* and show, using historical data and simulations, that these are significantly less volatile than the funding fee/funding rate for perpetual futures traded at popular venues. Margin closeouts on exchanges that offer perpetual futures are an analogue of liquidations on decentralized lending platforms [3]. We derive *implied maintenance margin* for a lending contract that can be contrasted with futures markets.

Loan contracts in this market are similar to traditional finance (TradFi) stock loans and can be studied through the lens of American options, [12, 9, 11]. A novel feature of these markets, in comparison to stock loans, is that debt positions which are not sufficiently collateralised are auctioned off to liquidators at a discount. This means that the holder of the loan effectively holds an American perpetual barrier option with a barrier above the strike.

In section 4 we empirically show that loan positions generally have a smaller and less volatile funding fee¹. Indeed, in Figure 1, we approximate the distribution of funding fees using 10000 simulations under three volatility regimes (see section 4 for more details on our model assumptions).

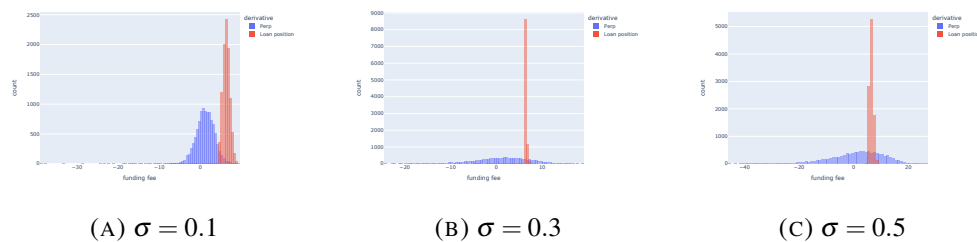


FIGURE 1. Funding fees distribution for Perps and Loan positions for three volatility regimes

The article is structured as follows. In section 2, we show that loan position on decentralised lending platforms, such as AAVE or Compound, is equivalent to entering perpetual American Barrier Option. In Section 3, we compare PnLs of loan positions and perpetual futures contracts. In Sections 4 and 5, we present a thorough empirical and simulation-based study comparing perpetual futures and lending positions under varied market conditions.

2. LEVERAGED TRADING VIA LENDING PLATFORMS

In this section, we show that entering a lending position is equivalent to entering a perpetual American Barrier option. We study the Profit-and-Loss (PnL) of this position, derive the implied funding fee/rate, and contrast these to perpetual futures contracts. Furthermore we derive maintenance margin rule for lending platform which is equivalent liquidation mechanism on decentralised lending platforms.

¹Simtopia.ai developed the dashboard allowing users to evaluate differences between perps and loan positions under varied market conditions: [ADD LINK](https://simtopia.ai)

2.1. A Perpetual American Barrier Option perspective on lending. Let $(P_t)_{t \geq 0}$ denote the price process of a risky asset, which we set to be ETH, with a dollar stablecoin being the numéraire. We take that stablecoin to be DAI, so for all $t \geq 0$, $1\text{ETH} = P_t\text{DAI}$ ².

Let $(r^{b,E}, r^{c,E}), (r^{b,D}, r^{c,D})$ be interest rates for borrowing and providing collateral for ETH and DAI respectively. Furthermore, Let $\theta^0 \in [0, \text{maxLTV})$ be an initial Loan-to-Value. Loan-to-Value corresponds to the initial haircut on the collateral which is less or equal to the maximal loan-to-value allowed by a protocol, meaning one borrows $\theta^0\text{ETH}$ of value for every unit of ETH deposited as collateral. To open a long-ETH loan position an agent 1) purchases 1 ETH on the market for P_0 of DAI, 2) deposits 1 ETH as collateral, 3) borrow $\theta^0 P_0$ of DAI against the collateral. We see that effectively only $P_0(1 - \theta^0)$ is required to establish the position. For capital efficiency, the agent may use a flashswap³ (or a flashloan and a swap) along the following steps:

- (1) Begin with $P_0(1 - \theta^0)$ of DAI
- (2) Obtain 1 ETH using flashswap (need to deposit P_0 DAI within one block for this to materialise)⁴
- (3) Deposit 1 ETH as collateral and start earning interests according to $e^{r^{c,E}t}$
- (4) Borrow $\theta^0 P_0$ of DAI against the collateral and start paying interests according to $\theta^0 P_0 e^{r^{b,D}t}$
- (5) Put together $\theta^0 P_0$ and initial amount $P_0(1 - \theta^0)$ of DAI to complete the flash-swap.

At any time the holder of the position may choose to pay back the loan $\theta^0 P_0 e^{r^{b,D}t}$ in exchange for the collateral with value $P_t e^{r^{c,E}t}$. Note that a rational agent will only do that if $\theta^0 P_0 e^{r^{b,D}t} \leq P_t e^{r^{c,E}t}$, otherwise it is better to walk away from the position. Hence, up until liquidation, the agent is entitled to the payoff

$$(P_t e^{r^{c,E}t} - \theta^0 P_0 e^{r^{b,D}t})_+ = e^{r^{b,D}t} (P_t e^{(r^{c,E} - r^{b,D})t} - \theta^0 P_0)_+,$$

where $x_+ = \max\{0, x\}$.

Let $\theta \in (\theta^0, 1)$ be a liquidation threshold. If the value of the asset falls too low, the position will be liquidated. This occurs at time τ^B , defined as

$$\tau^B := \inf \left\{ t \geq 0 : \theta P_t e^{r^{c,E}t} \leq \theta^0 P_0 e^{r^{b,D}t} \right\} = \inf \left\{ t \geq 0 : P_t e^{(r^{c,E} - r^{b,D})t} \leq \theta^{-1} \theta^0 P_0 \right\}.$$

The payoff accounting for liquidations is given by

$$\psi(P_t) := (P_t e^{r^{c,E}t} - \theta^0 P_0 e^{r^{b,D}t}) \mathbb{1}_{\{t < \tau^B\}}(t), \quad (1)$$

where $\mathbb{1}_{\{t < \tau^B\}}(t) = 1$ when $t < \tau^B$ and $\mathbb{1}_{\{t < \tau^B\}}(t) = 0$ when $t \geq \tau^B$. Note that $\psi(P_0) = P_0(1 - \theta^0)$. We can, therefore, consider this contract as equivalent to a down-and-out barrier option, where the position becomes worthless to its holder when the value of collateral falls sufficiently low. Note that taking a short loan position, that is, depositing DAI and borrowing ETH, does not lead to the payoff $-\psi(P_t)$. That means that, unlike in the futures markets, closing a position is not equivalent to entering an opposite trade.

²This is a standard accounting convention that does not take into account market frictions

³This takes advantage of the atomicity of the transactions executed on the chain but means the agent is exposed to slippage on a DEX of choice. When using a standard loan the ETH can be purchased on a centralised, potentially more liquid, exchange.

⁴One can also get DAI via flashloan and swap it for ETH. We ignore gas fee and slippage in the derivations.

Remark 1 (Flashswap close out). *To gain capital efficiency we used a flashswap to enter the loan position. To remain capital efficient we also require a flashswap to close the position. Indeed to pay off the loan an agent 1) obtains $\theta^0 P_0 e^{r^{b,D}t}$ using flashswap, 2) redeem $e^{r^{c,E}t}$ of ETH valued $P_t e^{r^{c,E}t}$ in DAI, 3) completes the flashswap and keeps the remaining part of the collateral.*

2.2. Loan Positions vs Perpetual Futures. We have seen in the previous section that to establish a long loan position with payoff (1) an agent requires $\psi(P_0)$ of DAI. The PnL of such a position is given by

$$\text{PnL}_t = \psi(P_t) - \psi(P_0) = (P_t e^{r^{c,E}t} - \theta^0 P_0 e^{r^{b,D}t}) \mathbb{1}_{\{t < \tau^B\}} - P_0(1 - \theta^0) \quad (2)$$

The PnL is driven by the change in the price of ETH/DAI and interest rates. Note that up to a liquidation event i.e for $t < \tau^B$ and for a short time t , which renders interest rates negligible, the $\text{PnL}_t \approx P_t - P_0$ but the agent only needs $P_0(1 - \theta^0)$ to enter the contract (and not P_0 which would be required to enter a spot trade). Hence we see that the initial LTV translates into leverage

$$\text{Leverage} = \frac{1}{1 - \text{initial LTV}} = \frac{1}{1 - \theta^0}. \quad (3)$$

The closer initial LTV θ^0 is to the liquidation threshold, the higher the leverage but also the risk of a liquidation.

2.3. Funding fee implied by a loan position. To compare the PnL of a loan position to the PnL of perpetual futures (cf. section 3) we introduce the notion of *implied funding fee/rate*. We can write for $t < \tau^B$

$$\begin{aligned} \text{PnL}_t &= \psi(P_t) - \psi(P_0) = P_t - P_0 - (P_t - P_0 - \psi(P_t) + \psi(P_0)) \\ &= P_t - P_0 - \text{IFF}_t, \quad \text{IFF}_t := \left(P_t(1 - e^{r^{c,E}t}) - \theta^0 P_0(1 - e^{r^{b,D}t}) \right) \end{aligned} \quad (4)$$

This shows one can decompose the PnL of the loan position into $P_t - P_0$, which corresponds to gains from leveraged trading and implied funding fee IFF_t , which is a counterpart of funding fee for perpetual futures before the liquidation time. Unlike for perpetual futures, the implied funding fee depends on the leverage, see (3), interest rates on a lending platform and price P_0 at which a position was established.

One can also define implied funding rate, IFR_t defined below

$$\text{IFF}_t = \int_0^t \text{IFR}_t dt, \quad \text{IFR}_t := \frac{d}{dt} \text{IFF}_t. \quad (5)$$

2.4. Trading on margins via Lending platform. Liquidations on decentralised lending platforms are an analogue of margin closeouts on exchanges that offer perpetual futures. On these exchanges to open the position an agent deposits funding on margin account, an initial margin, and the position is automatically closed out when for the first time

$$\text{initial margin} + \text{PnL}_t \leq \text{maintenance margin}_t,$$

where the maintenance margin is set by the exchange, see section (3.1) for more details. On decentralised lending platforms the liquidation happens when for the first time, for some $\theta \in (\theta^0, 1)$,

$$\theta P_t e^{r^{c,E}t} \leq \theta^0 P_0 e^{r^{b,D}t}.$$

By rearranging this inequality, we obtain

$$\underbrace{\psi(P_0)}_{\text{initial margin}} + \underbrace{\psi(P_t) - \psi(P_0)}_{\text{PnL}_t} = \psi(P_t) = P_t e^{r^{c,E}t} - \theta^0 P_0 e^{r^{b,D}t} \leq \underbrace{(1 - \theta) P_t e^{r^{c,E}t}}_{\text{maintenance margin}}.$$

Hence we see that liquidations on decentralised lending platform correspond to margin closeouts with the maintenance margin given by $(1 - \theta)P_t e^{r^{c,E}t}$. In other words one can think of $(1 - \theta)e^{r^{c,E}t}$ to be maintenance margin parameter implied by a lending platform.

2.5. Impact of lending on spot markets. The funding fee for perpetual futures contracts is designed to incentivise trades that tend to narrow the futures-spot gap and impact the spot market. Next, we briefly describe what impact lending protocols may have on the spot market.

On the one hand, entering a long ETH loan position requires a swap of DAI for ETH, which is expected to increase the price of ETH/DAI. On the other hand, opening a long ETH loan position, *ceteris paribus*, decreases utilisation and hence interest rate for lending ETH, and increases utilisation and therefore interest rate for borrowing DAI (how these rates change depends on the elasticity of market supply and demand curves and interest rate mechanism, see [4] for more details), which has a negative impact on the PnL of the position and also increases liquidation risk.

What is more, if liquidation happens, the liquidator, by taking over a delinquent position, will be in possession of a risky asset (redeemed ETH) and want to exchange it for DAI (to mitigate adverse selection). Consequently, *ceteris paribus*, the supply of ETH on the external market increases, pushing the price of ETH down (adversarial liquidation and liquidation spirals will further accelerate the above scenario, see [3] for more details).

Finally, if the interest rate on ETH is falling, yield-seeking agents may start exiting their positions and seek to change ETH for DAI, creating further pressure on ETH price to go down.

We study some of these effects by performing simulations under varying market conditions in Section 4.

3. PERPETUAL FUTURES

An agent seeking to enter a leveraged long or short trading position instead of using loan protocol may use a venue offering perpetual future contract. We recall a definition of such a contract, stated in [10]

Definition 1 (Perpetual Futures). *A perpetual contract with price F and a funding rate designed to track spot price P is an agreement between two parties referred to as the long side and short side. There is 0 cost to enter the agreement. The long (short) side has the right to terminate the contract at any time $t \geq 0$, at which point they will receive a payment of $F_t - F_0$ ($-(F_t - F_0)$). In return, for a fixed scaling parameter $\kappa > 0$, at every t_i , $i = 1, 2, \dots$ the long side must pay to the short side the $\frac{\kappa}{t_{i+1} - t_i} \int_{t_i}^{t_{i+1}} (F_s - P_s) ds$ referred to as the funding rate up until the contract is terminated.*

As the frequency of funding rate payment increases, the funding rate converges to $\kappa \int_0^t (F_s - P_s) ds$. The funding fee paid in numéraire and its value at time t is given by $\int_0^t e^{r(t-s)} (F_s - P_s) ds$ where, $r > 0$ is a risk-free rate⁵

The PnL generated by the long perpetual futures is given by

$$\text{PnL}_t^F := F_t - F_0 - \kappa \int_0^t e^{r(t-s)} (F_s - P_s) ds.$$

⁵For simplicity we work with a single risk-free rate, ignoring the fact that rates for lending and borrowing may differ. We refer interested readers to [6] for an overview of the mathematical theory needed to study the case of different rates for lending and borrowing

The fair price of the perpetual futures can be deduced by requiring that there is no cash-and-carry arbitrage opportunities in the market. An agent who begins with zero capital, shorts spot and at the same goes long futures contracts (or goes long spot and short futures contract) shouldn't at any time have cash flows which are positive with probability one.

Through the rest of the section we set $r = r^{c,D}$. The economic rationale for this is that $r^{c,D}$ is a rate at which blockchain users are willing to lend liquid stablecoins, and hence we can take it as a proxy for risk-free rates on these markets.

We get a similar result to [10, 8] on how perpetual future and underlying price and that scaling parameter κ is fixed are related.

Lemma 1. *Assume that spot and futures prices are given by stochastic process $(P_t)_{t \geq 0}$ and $(S_t)_{t \geq 0}$ adapted to a natural filtration $(\mathcal{F}_t)_{t \geq 0}$ and that scaling parameter κ is fixed. Assume there is a risk neutral measure $\mathbb{Q}$ under which the underlying asset is a martingale, and that*

$$\mathbb{E}^{\mathbb{Q}} \left[\int_0^\infty e^{-r^{c,D}s} (|P_s| + |F_s|) ds \right] < \infty. \quad (6)$$

Then either there is arbitrage or the following relation holds

$$F_t = P_t \left(1 + \frac{r^{c,D}}{\kappa} \right) \quad \text{for all } t > 0.$$

Proof. Consider the wealth process Z of an agent who

- (i) begins with zero capital⁶,
- (ii) initially borrows a quantity P_0 at the risk-free interest rate, to short the spot
- (iii) enters a long perpetual contract (the cost of doing which is zero⁷)

Throughout $r = r^{c,D}$. The dynamic of Z_t is given by

$$Z_t = \underbrace{P_0 e^{rt} - P_t}_{\text{PnL from short spot}} + \underbrace{F_t - F_0 - \kappa \int_0^t e^{r(t-s)} (F_s - P_s) ds}_{\text{PnL from long futures}}.$$

Let $X_t = e^{-rt} (F_t - P_t)$. Then

$$e^{-rt} Z_t = X_t - (e^{-rt} F_0 - P_0) - \kappa \int_0^t X_s ds.$$

Lack of arbitrage requires that $(e^{-rt} Z_t)_t$ is a martingale and hence

$$0 = \mathbb{E}^{\mathbb{Q}} \left[X_t - (e^{-rt} F_0 - P_0) - \kappa \int_0^t X_s ds \right].$$

By rearranging the terms we have

$$d\mathbb{E}^{\mathbb{Q}}[X_t] = -re^{-rt} F_0 + \kappa \mathbb{E}^{\mathbb{Q}}[X_t].$$

Observe that⁸

$$d(e^{-\kappa t} \mathbb{E}^{\mathbb{Q}}[X_t]) = -re^{-(\kappa+r)t} F_0 dt$$

Integrating both side from 0 to t we arrive at

$$\mathbb{E}^{\mathbb{Q}}[X_t] = e^{\kappa t} X_0 + \frac{r}{\kappa + r} F_0 e^{\kappa t} (e^{-(\kappa+r)t} - 1) = e^{\kappa t} \left(F_0 \frac{1}{1 + \frac{r}{\kappa}} - P_0 \right) + \frac{r}{\kappa + r} F_0 e^{-rt}.$$

⁶This is simply to avoid having to account for the interest they should earn on their initial capital.

⁷In practice perpetual futures are being traded on margins and agents need to deposit a certain amount of collateral on the margin account.

⁸For finite expiry futures we would have that at expiration $F_T = P_T$, giving a terminal condition $\mathbb{E}^{\mathbb{Q}}[X_T] = 0$ from which we could deduce non-arbitrage price F_0 .

Note that

$$\int_0^t e^{\kappa s} ds \left| F_0 \frac{1}{1 + \frac{r}{\kappa}} - P_0 \right| \leq \int_0^t |\mathbb{E}^\mathbb{Q}[X_s]| ds + \int_0^t \frac{r}{\kappa + r} F_0 e^{-rs} ds.$$

Using Jensen's inequality

$$\lim_{t \rightarrow \infty} \frac{1}{\kappa} (e^{\kappa t} - 1) \left| F_0 \frac{1}{1 + \frac{r}{\kappa}} - P_0 \right| \leq \mathbb{E}^\mathbb{Q} \left[\int_0^\infty |X_s| ds \right] + \frac{r}{(\kappa + r)r} F_0.$$

Assumption (6) implies that

$$\lim_{t \rightarrow \infty} \frac{1}{\kappa} (e^{\kappa t} - 1) \left| F_0 \frac{1}{1 + \frac{r}{\kappa}} - P_0 \right| < \infty.$$

Since $\kappa > 0$, this is only possible when

$$F_0 = P_0 \left(1 + \frac{r}{\kappa} \right).$$

□

3.1. Margin considerations. To compare perpetual futures to the corresponding loan position, assume that there is a venue that allows agents to take long ⁹ perpetual future position using the same leverage as a lending platform, that is $\frac{1}{1-\theta^0}$, see (3).

We first assume that the no-arbitrage condition established in Lemma 1 holds, that is $F_t = P_t(1 + \frac{r}{\kappa})$. That means that one ETH on a spot market valued P_t corresponds to $(1 + \frac{r}{\kappa})^{-1} F_t$ on a perpetual futures market.

To enter a long perpetual futures position, an agent

- (1) Deposits $P_0(1 - \theta^0)$ on margin account. This corresponds to the initial margin.
- (2) Gains exposure to $P_0 = F_0(1 + \frac{r}{\kappa})^{-1}$ perpetual futures with a cash-flow

$$\left(1 + \frac{r}{\kappa} \right)^{-1} \left(F_t - F_0 - \kappa \int_0^t e^{r(t-s)} (F_s - P_s) ds \right). \quad (7)$$

Recall that θ^0 is the initial LTV which is less or equal than the maximal loan-to-value, max LTV, allowed by the protocol. We assume that¹⁰

$$\text{maintenance margin}_t := P_t \cdot \theta^F, \quad 0 \leq \theta^F \leq 1.$$

The position gets liquidated when for the first time

$$\text{initial margin} + \text{PnL}_t^F \leq \text{maintenance margin}_t.$$

This happens at the stopping time τ^F given by

$$\tau^F := \inf_t \{ t \geq 0 : P_t \cdot \theta^F \geq P_0(1 - \theta^0) + \text{PnL}_t^F \}.$$

When the position gets liquidated, the agent loses her initial margin.

4. SIMULATION EXPERIMENTS

In this section we present numerical experiments to compare the PnL, defined in (7), the funding fee and liquidation times of the Perpetual Futures (Perps) with the PnL, defined in (2), implied funding fee, defined in (4), and liquidation time for the loan position under various market conditions.

We make the following modelling assumptions:

- i) ETH price process follows a geometric Brownian motion

$$dP_t = \mu P_t dt + \sigma P_t dW_t, \quad P_0 = p_0.$$

⁹the short position works exactly the same

¹⁰This corresponds to the convention on Binance or Vega protocol

- ii) It has been observed in [10], that the no-arbitrage condition $F_t = P_t(1 + \frac{r}{\kappa})$ is often violated in practice. This is because we have not considered market frictions such as slippage, gas fees etc. To reflect market data, see [10], we model the price of a perpetual future $(F_t)_{t \geq 0}$ by the mean-reverting process

$$dF_t = \lambda(P_t - F_t)dt + \sigma^F dW_t^F, \quad F_0 = P_0 \left(1 + \frac{r^{c,D}}{\kappa}\right),$$

where $[W^F, W](t) = \rho t$.

- iii) We model the impact of the lending on the spot described in Section 2.5 as follows. The utilisation of the DAI and the ETH pool, denoted by U_t^{ABC} , $ABC \in \{DAI, ETH\}$, is modelled in terms of the ETH price P_t as

$$U_t^{ABC} = g(v_t^{ABC}), \quad \text{for } t \geq 0$$

$$\text{where } dv_t^{ABC} = \alpha^{ABC} dP_t, \quad g(x) = \frac{1}{1 + e^{-x}}.$$

In other words α^{ABC} measures relative change of v with respect to change of P and use sigmoid function g to map v to the utilisation $U \in [0, 1]$. The choice of g is arbitrary and can be calibrated to data, we use the sigmoid function as it is convenient to bring any real value to the interval $(0, 1)$. When α^{ETH} is negative and α^{DAI} is positive, an increase in the price of ETH decreases the utilisation of ETH and increases the utilisation of DAI, as speculated in Section 2.5. The absolute values of $\alpha^{ETH}, \alpha^{DAI}$ provide the magnitude of this association.

- iv) The interest rates r_t^{ABC} , $ABC \in \{DAI, ETH\}$, are calculated using the piece-wise linear interest rate model depending on U_t^{ABC} given by

$$r^{ABC}(U^{ABC}) = r_0 + r_1 \cdot \frac{U^{ABC}}{U^*} \cdot \mathbf{1}_{U^{ABC} \leq U^*} + \left(r_1 + r_2 \cdot \frac{U^{ABC} - U^*}{1 - U^*} \right) \cdot \mathbf{1}_{U^{ABC} > U^*}$$

We fix the parameters for

- the ETH price process: we make simulations for $\mu \in \{-0.3, 0.0, 0.3\}$, $\sigma \in \{0.1, 0.3, 0.5\}$, $p_0 = 2000$,
 - the Perps price process: $\lambda = 100$, $\sigma^F = 100$, $\kappa = 1$, $\rho = 0$,
 - the interest rate model: $\alpha^{ETH} = -0.15$, $\alpha^{DAI} = 0.05$, $U^* = 0.45$, $r_0 = 0$, $r_1 = 0.04$, $r_2 = 2.5$,¹¹
 - the loan position: initial LTV $\theta^0 = 0.75$, loan liquidation threshold $\theta = 0.85$,¹²
 - the Perps position: maintenance margin parameter $\theta^F = 0.0875$,¹³.
- With this choice of parameters, the agent enters a 4x leveraged position and has a maintenance margin requirement of 8.75%.
- the number of Monte Carlo simulations: 10000.

The simulations are run for a period of one year, and we plot the empirical distributions of PnL, (implied) funding fee and liquidation times under different market scenarios provided by μ, σ .

Figure 2 provides simulated samples of the ETH and the Perps prices under the above models and parameters

¹¹parameter values taken from <https://docs.aave.com/risk/liquidity-risk/borrow-interest-rate>

¹²Risk parameter values from some currencies in Aave v3 and Morpho - Aave v3

¹³We use the maintenance margin found in Binance or dYdX, e.g. <https://help.dydx.exchange/en/articles/4800587-12-perpetual-contract-specs>

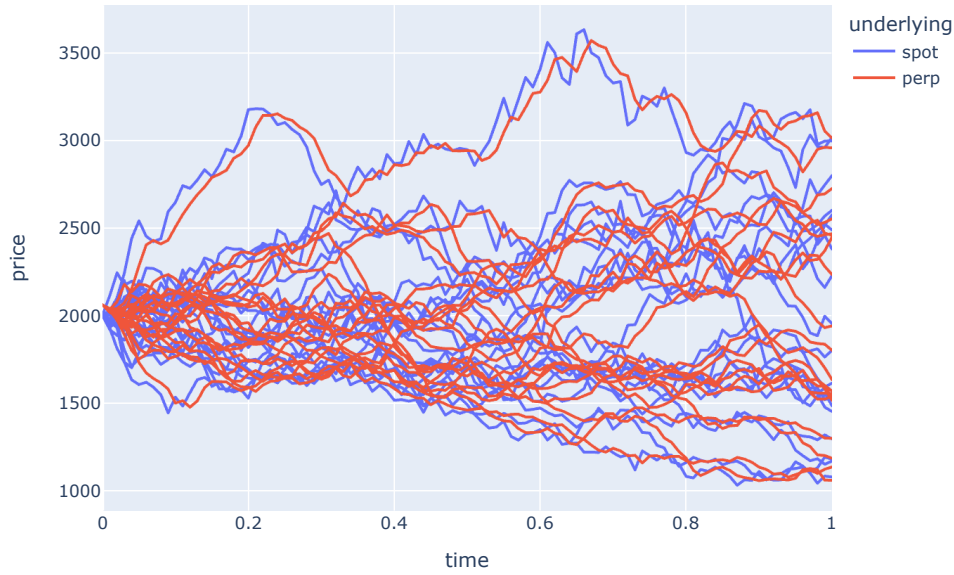


FIGURE 2. Simulated samples of ETH and perps price processes for $\mu = 0, \sigma = 0.3$

Figure 3 provides the PnL for perps and loan positions for fixed $\mu = 0$ and $\sigma \in \{0.1, 0.3, 0.5\}$ at time $t=0.25$. Taking into account the empirical CDF in Figure 6 in those same scenarios, our experiments show that Loan positions are typically liquidated earlier than Perps, as CEX allow for a small maintenance margin (in our simulations $\theta^F = 0.05$) giving more margin before liquidation compared to Decentralised Liquidation Protocols (in our simulations $\theta = 0.95$). Nevertheless, our simulations yield similar PnL for Perps and Loan positions, with Loan positions having a much less volatile funding fee (see Figure 5). Note that as the volatility σ increases, liquidations increase too, hence the multi-modal shape of the PnL distributions in Figures 3, 4.

Figure 5 shows that funding fees for the loan positions are, on average, lower than funding fees for Perps and also have less volatility.

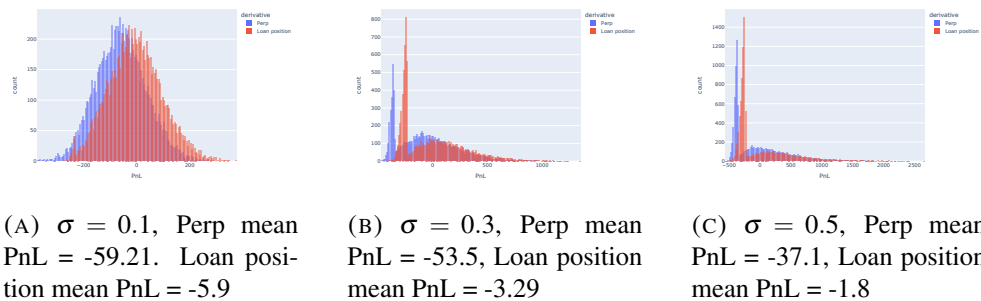


FIGURE 3. PnL for Perps and Loan positions at time $t = 0.25$ (after 3 months) for fixed $\mu = 0$. and $\sigma \in \{0.1, 0.3, 0.5\}$

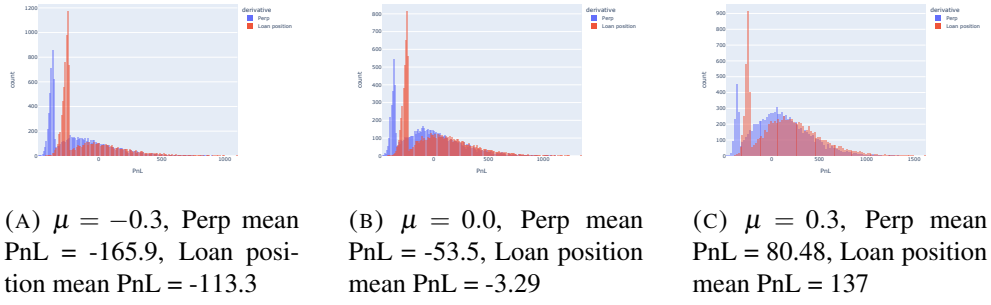


FIGURE 4. PnL for Perps and Loan positions at time $t = 0.25$ (after 3 months) for fixed $\sigma = 0.3$ and $\mu \in \{-0.3, 0., 0.3\}$

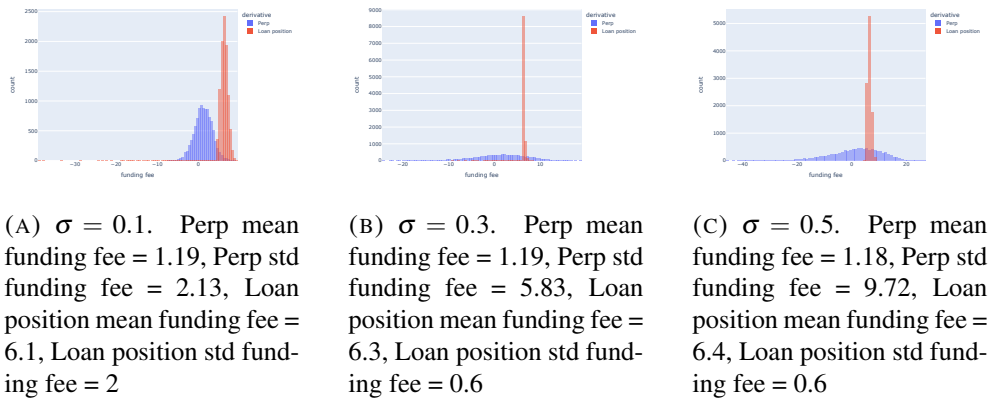


FIGURE 5. Funding fees distribution for Perps and Loan positions for fixed $\mu = 0.0$ and $\sigma \in \{0.1, 0.3, 0.5\}$

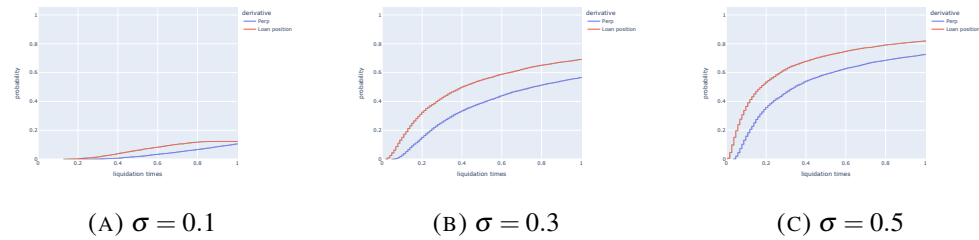


FIGURE 6. CDF of liquidation times for Perps and Loan positions for fixed $\mu = 0.0$ and $\sigma \in \{0.1, 0.3, 0.5\}$

5. EMPIRICAL DATA ANALYSES

5.1. Data. We use CoinGlass API (coinglass.com) to fetch time series of 8-hour funding rates of USD-margined perps from various exchanges, as well as the corresponding time series of the risk asset price denominated in USD.

We collect the historical data on different crypto assets' lending and borrowing rates on Aave with GraphQL API from The Graph (thegraph.com).

Our dataset on funding rates contains data points at 8-hour intervals spanning from 8:00 on 16 April 2023 to 8:00 on 13 October 2023.

The exchanges we use to compare with the decentralised finance (DeFi) loan position include Binance, Bybit, OKX, Huobi, Gate, Bitget, dYdX and CoinEx.

The crypto assets we look into in our simulation are BTC, ETH, LINK, UNI, AAVE, CRV, MKR and SNX.

5.2. Implied funding fee/rate. To make comparisons with other platforms, we first calculate implied funding rates, using discretization of (5) and historical data.

Particularly, we assume the lending platform user is taking a DAI-margined long position for some risk asset. The leverage ratio is set to be 5x.¹⁴

For the implied funding rate calculation, we use the USD-dominated price series of the risk asset fetched from CoinGlass, and interpolate DAI's borrow interest rates and the risk asset's supply (deposit) interest rates on Aave at 8-hour intervals based on the data collected from The Graph.

5.3. Comparison of funding rates between platforms. In Figure 7, we superimpose loan contract 8-hour implied funding rate series on the historical funding rates of three crypto exchanges: dYdX, Binance and Huobi. Implied funding rates appear less volatile, and their absolute values are almost always lower than those on other platforms.

¹⁴For comparability purposes, we set the leverage to be the same across all crypto-assets. In reality, it is difficult for low-liquidity crypto-assets to achieve high leverage.

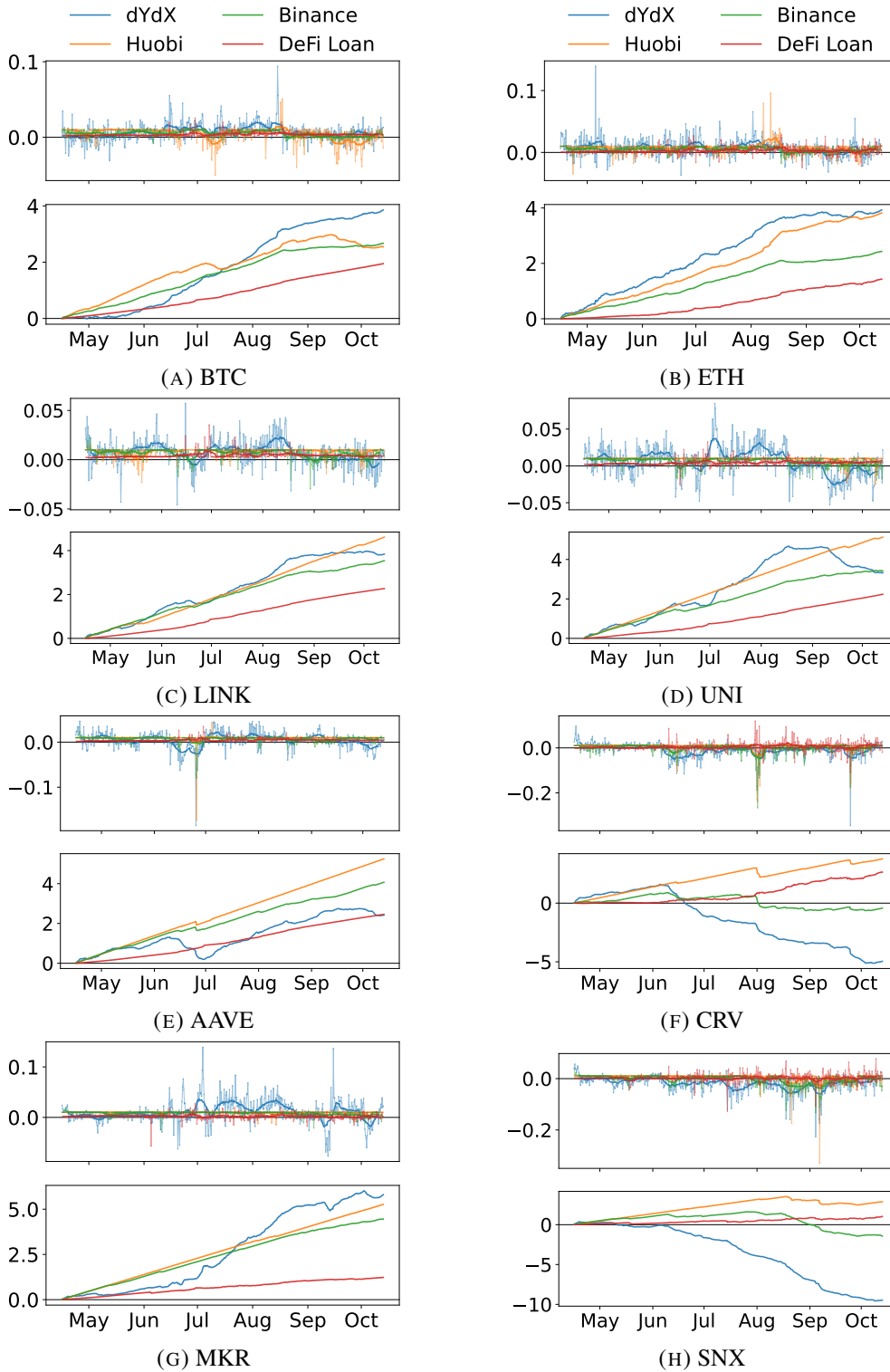


FIGURE 7. In the upper plot for each crypto asset we have implied funding rates, calculated as (5) with historical data (see configurations in §5.2) compared with historical funding rates of dYdX, Huobi and Binance from April to October 2023. The semi-transparent lines with markers are 8-hour data points; the non-transparent lines represent 7-day moving average of the 8-hour funding rates. In the lower plot, we have implied cumulative funding fee, calculated as (4), is compared with that of the respective exchange.

(A) BTC

	std	mean	min	25%	50%	75%	max	count
DeFi Loan	0.0026	0.0036	-0.0011	0.0023	0.0028	0.0038	0.0255	540
Binance	0.0043	0.0049	-0.0109	0.0021	0.0049	0.0096	0.0100	540
Bybit	0.0044	0.0059	-0.0086	0.0024	0.0071	0.0100	0.0100	540
Bitget	0.0046	0.0050	-0.0103	0.0021	0.0054	0.0093	0.0109	540
OKX	0.0070	0.0028	-0.0331	-0.0017	0.0024	0.0069	0.0301	540
Gate	0.0071	0.0016	-0.0315	-0.0030	0.0025	0.0073	0.0100	540
Huobi	0.0096	0.0047	-0.0501	0.0018	0.0100	0.0100	0.0507	539
dYdX	0.0114	0.0071	-0.0309	0.0007	0.0067	0.0123	0.0939	540
CoinEx	0.0210	-0.0036	-0.1430	-0.0156	-0.0024	0.0099	0.0718	540

(B) ETH

	std	mean	min	25%	50%	75%	max	count
Binance	0.0045	0.0045	-0.0177	0.0014	0.0046	0.0087	0.0100	540
Bybit	0.0050	0.0054	-0.0126	0.0018	0.0067	0.0100	0.0100	540
Bitget	0.0051	0.0044	-0.0177	0.0017	0.0048	0.0090	0.0109	540
DeFi Loan	0.0058	0.0027	-0.0245	0.0003	0.0017	0.0043	0.0284	540
Gate	0.0059	0.0021	-0.0221	-0.0020	0.0027	0.0069	0.0100	540
OKX	0.0079	0.0035	-0.0500	-0.0012	0.0033	0.0083	0.0274	540
Huobi	0.0088	0.0070	-0.0346	0.0044	0.0100	0.0100	0.0959	540
dYdX	0.0145	0.0072	-0.0367	-0.0008	0.0071	0.0137	0.1391	540
CoinEx	0.0159	0.0000	-0.0692	-0.0085	0.0002	0.0095	0.0759	540

(C) LINK

	std	mean	min	25%	50%	75%	max	count
DeFi Loan	0.0031	0.0042	0.0016	0.0030	0.0034	0.0041	0.0354	540
Huobi	0.0042	0.0085	-0.0231	0.0100	0.0100	0.0100	0.0150	539
Binance	0.0052	0.0065	-0.0295	0.0041	0.0093	0.0100	0.0222	540
Bitget	0.0053	0.0066	-0.0299	0.0044	0.0092	0.0100	0.0230	540
Gate	0.0054	0.0027	-0.0315	-0.0008	0.0032	0.0068	0.0100	540
Bybit	0.0065	0.0065	-0.0600	0.0040	0.0100	0.0100	0.0285	540
OKX	0.0109	0.0029	-0.0603	-0.0035	0.0030	0.0094	0.0506	540
dYdX	0.0124	0.0070	-0.0457	0.0012	0.0075	0.0129	0.0571	540
CoinEx	0.0202	-0.0058	-0.0873	-0.0155	-0.0037	0.0067	0.0620	540

(D) UNI

	std	mean	min	25%	50%	75%	max	count
Huobi	0.0033	0.0095	-0.0247	0.0100	0.0100	0.0100	0.0100	540
DeFi Loan	0.0038	0.0041	-0.0256	0.0023	0.0032	0.0050	0.0328	540
Gate	0.0055	0.0058	-0.0211	0.0027	0.0088	0.0100	0.0100	540
Binance	0.0057	0.0063	-0.0192	0.0044	0.0100	0.0100	0.0100	540
Bitget	0.0058	0.0064	-0.0197	0.0045	0.0092	0.0100	0.0109	540
Bybit	0.0061	0.0062	-0.0280	0.0034	0.0100	0.0100	0.0197	540
OKX	0.0109	0.0043	-0.0359	-0.0021	0.0048	0.0115	0.0313	540
CoinEx	0.0172	-0.0076	-0.0729	-0.0175	-0.0066	0.0044	0.0478	540
dYdX	0.0197	0.0061	-0.0527	-0.0037	0.0070	0.0174	0.0848	540

(E) AAVE

	std	mean	min	25%	50%	75%	max	count
DeFi Loan	0.0032	0.0045	0.0018	0.0031	0.0037	0.0046	0.0349	540
Gate	0.0065	0.0087	-0.1087	0.0100	0.0100	0.0100	0.0100	540
Binance	0.0070	0.0075	-0.0797	0.0079	0.0100	0.0100	0.0240	540
Bitget	0.0070	0.0075	-0.0804	0.0079	0.0095	0.0101	0.0239	540
Huobi	0.0081	0.0097	-0.1738	0.0100	0.0100	0.0100	0.0437	538
Bybit	0.0100	0.0055	-0.1242	0.0027	0.0100	0.0100	0.0383	540
OKX	0.0153	0.0032	-0.1650	-0.0036	0.0037	0.0121	0.0666	540
dYdX	0.0182	0.0044	-0.1851	-0.0031	0.0056	0.0135	0.0469	540
CoinEx	0.0381	-0.0030	-0.3564	-0.0171	-0.0025	0.0100	0.5553	540

(F) CRV

	std	mean	min	25%	50%	75%	max	count
DeFi Loan	0.0196	0.0049	-0.1086	-0.0029	0.0020	0.0110	0.1188	540
Bybit	0.0206	0.0027	-0.2381	0.0080	0.0100	0.0100	0.0217	540
Gate	0.0210	0.0046	-0.3219	0.0086	0.0100	0.0100	0.0100	540
Huobi	0.0212	0.0070	-0.2391	0.0100	0.0100	0.0100	0.0100	539
Binance	0.0232	-0.0008	-0.2666	-0.0036	0.0100	0.0100	0.0100	540
Bitget	0.0234	-0.0009	-0.2706	-0.0045	0.0092	0.0100	0.0109	540
dYdX	0.0308	-0.0093	-0.3450	-0.0190	-0.0051	0.0080	0.0964	540
OKX	0.0424	-0.0079	-0.6419	-0.0197	-0.0050	0.0107	0.0977	540
CoinEx	0.0606	-0.0205	-0.7500	-0.0362	-0.0166	0.0017	0.4581	540

(G) MKR

	std	mean	min	25%	50%	75%	max	count
Huobi	0.0023	0.0097	-0.0143	0.0100	0.0100	0.0100	0.0100	540
Binance	0.0041	0.0082	-0.0249	0.0089	0.0100	0.0100	0.0178	540
Bitget	0.0043	0.0082	-0.0255	0.0090	0.0097	0.0103	0.0175	540
DeFi Loan	0.0053	0.0023	-0.0569	0.0008	0.0026	0.0034	0.0270	540
Gate	0.0056	0.0073	-0.0320	0.0079	0.0100	0.0100	0.0100	540
Bybit	0.0063	0.0076	-0.0546	0.0100	0.0100	0.0100	0.0200	540
OKX	0.0175	0.0025	-0.0915	-0.0082	0.0023	0.0127	0.0745	540
dYdX	0.0223	0.0107	-0.0766	0.0011	0.0085	0.0192	0.1390	540
CoinEx	0.0324	-0.0068	-0.2696	-0.0246	-0.0062	0.0100	0.1088	540

(H) SNX

	std	mean	min	25%	50%	75%	max	count
Gate	0.0138	0.0052	-0.1342	0.0100	0.0100	0.0100	0.0100	540
DeFi Loan	0.0169	0.0019	-0.1276	-0.0031	0.0024	0.0068	0.0774	540
Binance	0.0195	-0.0027	-0.1322	-0.0092	0.0064	0.0100	0.0233	540
Bitget	0.0196	-0.0028	-0.1325	-0.0092	0.0063	0.0099	0.0231	540
Bybit	0.0199	-0.0012	-0.1815	-0.0070	0.0100	0.0100	0.0524	540
Huobi	0.0201	0.0053	-0.3293	0.0100	0.0100	0.0100	0.0100	538
OKX	0.0270	-0.0071	-0.2342	-0.0208	-0.0037	0.0102	0.0494	540
dYdX	0.0297	-0.0176	-0.2268	-0.0287	-0.0122	0.0011	0.0568	540
CoinEx	0.0332	-0.0164	-0.3270	-0.0326	-0.0127	0.0019	0.1391	540

TABLE 1. Summary statistics of 8-hour funding rates from April to October 2023 across various platforms. The DeFi loan position's implied funding rates are calculated as Eq. 5 in a simulated environment with historical data on price and interest rates (see configurations in §5.2); funding rates of other platforms are directly fetched historical values. The Comparison metrics include standard deviation (std), mean, minimum (min), first quartile (25%), medium (50%), third quartile (75%) and maximum (max).

Table 1 presents summary statistics of the (implied) funding rates on various platforms. In line with what is illustrated in Figure 7, the comparison results in Table 1 show that the DeFi loan position exhibits one of the lowest volatilities in funding rate among other platforms for long positions. This applies to all risky assets except for ETH and MRK, where the DeFi loan position is ranked fourth.

Table 2 exhibits the correlation matrix of funding rates between various platforms. All orderbook-based exchanges appear to be highly positively correlated in terms of funding rates in the majority of cases (with few exceptions such as Huobi and dYdX's correlation in terms of MKR's funding rates). Implied funding rates are little, and sometimes even slightly negatively correlated with the funding rates from those orderbook-based exchanges.

(A) BTC

	Binance	Bybit	OKX	Huobi	Gate	Bitget	dYdX	CoinEx	DeFi Loan
Binance	1.00	0.39	0.44	0.32	0.33	0.94	0.29	0.06	0.05
Bybit	0.39	1.00	0.32	0.35	0.43	0.37	0.23	0.17	-0.01
OKX	0.44	0.32	1.00	0.28	0.37	0.41	0.28	0.14	-0.01
Huobi	0.32	0.35	0.28	1.00	0.44	0.31	0.09	0.32	0.01
Gate	0.33	0.43	0.37	0.44	1.00	0.31	0.09	0.38	0.00
Bitget	0.94	0.37	0.41	0.31	0.31	1.00	0.27	0.05	0.05
dYdX	0.29	0.23	0.28	0.09	0.09	0.27	1.00	0.08	0.12
CoinEx	0.06	0.17	0.14	0.32	0.38	0.05	0.08	1.00	0.10
DeFi Loan	0.05	-0.01	-0.01	0.01	0.00	0.05	0.12	0.10	1.00

(B) ETH

	Binance	Bybit	OKX	Huobi	Gate	Bitget	dYdX	CoinEx	DeFi Loan
Binance	1.00	0.53	0.38	0.19	0.21	0.92	0.35	0.09	-0.13
Bybit	0.53	1.00	0.30	0.21	0.18	0.48	0.30	0.06	-0.15
OKX	0.38	0.30	1.00	0.39	0.54	0.36	0.11	0.40	0.03
Huobi	0.19	0.21	0.39	1.00	0.44	0.21	0.06	0.34	0.15
Gate	0.21	0.18	0.54	0.44	1.00	0.23	-0.01	0.46	0.22
Bitget	0.92	0.48	0.36	0.21	0.23	1.00	0.32	0.11	-0.10
dYdX	0.35	0.30	0.11	0.06	-0.01	0.32	1.00	0.01	-0.22
CoinEx	0.09	0.06	0.40	0.34	0.46	0.11	0.01	1.00	0.24
DeFi Loan	-0.13	-0.15	0.03	0.15	0.22	-0.10	-0.22	0.24	1.00

(C) LINK

	Binance	Bybit	OKX	Huobi	Gate	Bitget	dYdX	CoinEx	DeFi Loan
Binance	1.00	0.45	0.38	0.07	0.35	0.98	0.32	0.31	-0.02
Bybit	0.45	1.00	0.41	0.26	0.45	0.43	0.30	0.40	-0.01
OKX	0.38	0.41	1.00	0.12	0.33	0.35	0.23	0.36	-0.04
Huobi	0.07	0.26	0.12	1.00	0.18	0.06	0.06	0.12	0.09
Gate	0.35	0.45	0.33	0.18	1.00	0.35	0.22	0.45	-0.04
Bitget	0.98	0.43	0.35	0.06	0.35	1.00	0.32	0.29	-0.02
dYdX	0.32	0.30	0.23	0.06	0.22	0.32	1.00	0.22	0.03
CoinEx	0.31	0.40	0.36	0.12	0.45	0.29	0.22	1.00	-0.06
DeFi Loan	-0.02	-0.01	-0.04	0.09	-0.04	-0.02	0.03	-0.06	1.00

(D) UNI

	Binance	Bybit	OKX	Huobi	Gate	Bitget	dYdX	CoinEx	DeFi Loan
Binance	1.00	0.48	0.38	0.23	0.17	0.98	0.37	0.44	-0.17
Bybit	0.48	1.00	0.41	0.15	0.28	0.48	0.31	0.40	-0.01
OKX	0.38	0.41	1.00	0.11	0.19	0.37	0.28	0.40	-0.12
Huobi	0.23	0.15	0.11	1.00	0.01	0.23	0.14	0.09	-0.08
Gate	0.17	0.28	0.19	0.01	1.00	0.17	0.02	0.23	-0.02
Bitget	0.98	0.48	0.37	0.23	0.17	1.00	0.36	0.43	-0.17
dYdX	0.37	0.31	0.28	0.14	0.02	0.36	1.00	0.36	-0.18
CoinEx	0.44	0.40	0.40	0.09	0.23	0.43	0.36	1.00	-0.04
DeFi Loan	-0.17	-0.01	-0.12	-0.08	-0.02	-0.17	-0.18	-0.04	1.00

(E) AAVE

	Binance	Bybit	OKX	Huobi	Gate	Bitget	dYdX	CoinEx	DeFi Loan
Binance	1.00	0.77	0.60	0.55	0.65	1.00	0.49	0.28	-0.14
Bybit	0.77	1.00	0.60	0.56	0.62	0.77	0.42	0.28	-0.03
OKX	0.60	0.60	1.00	0.48	0.53	0.60	0.43	0.28	-0.07
Huobi	0.55	0.56	0.48	1.00	0.78	0.55	0.17	0.20	0.02
Gate	0.65	0.62	0.53	0.78	1.00	0.65	0.40	0.21	0.01
Bitget	1.00	0.77	0.60	0.55	0.65	1.00	0.49	0.28	-0.13
dYdX	0.49	0.42	0.43	0.17	0.40	0.49	1.00	0.22	-0.08
CoinEx	0.28	0.28	0.28	0.20	0.21	0.28	0.22	1.00	-0.09
DeFi Loan	-0.14	-0.03	-0.07	0.02	0.01	-0.13	-0.08	-0.09	1.00

(F) CRV

	Binance	Bybit	OKX	Huobi	Gate	Bitget	dYdX	CoinEx	DeFi Loan
Binance	1.00	0.76	0.72	0.73	0.77	1.00	0.59	0.56	-0.07
Bybit	0.76	1.00	0.71	0.77	0.78	0.76	0.53	0.52	0.04
OKX	0.72	0.71	1.00	0.71	0.79	0.72	0.49	0.61	-0.07
Huobi	0.73	0.77	0.71	1.00	0.87	0.73	0.46	0.53	0.04
Gate	0.77	0.78	0.79	0.87	1.00	0.77	0.46	0.58	0.01
Bitget	1.00	0.76	0.72	0.73	0.77	1.00	0.59	0.56	-0.06
dYdX	0.59	0.53	0.49	0.46	0.46	0.59	1.00	0.38	-0.08
CoinEx	0.56	0.52	0.61	0.53	0.58	0.56	0.38	1.00	0.02
DeFi Loan	-0.07	0.04	-0.07	0.04	0.01	-0.06	-0.08	0.02	1.00

(G) MKR

	Binance	Bybit	OKX	Huobi	Gate	Bitget	dYdX	CoinEx	DeFi Loan
Binance	1.00	0.49	0.37	0.05	0.33	0.99	0.21	0.34	-0.03
Bybit	0.49	1.00	0.31	0.01	0.33	0.50	0.28	0.21	0.01
OKX	0.37	0.31	1.00	0.07	0.29	0.37	0.19	0.34	0.07
Huobi	0.05	0.01	0.07	1.00	0.05	0.04	-0.06	0.06	0.05
Gate	0.33	0.33	0.29	0.05	1.00	0.35	0.10	0.24	0.19
Bitget	0.99	0.50	0.37	0.04	0.35	1.00	0.21	0.33	-0.02
dYdX	0.21	0.28	0.19	-0.06	0.10	0.21	1.00	0.07	-0.06
CoinEx	0.34	0.21	0.34	0.06	0.24	0.33	0.07	1.00	0.09
DeFi Loan	-0.03	0.01	0.07	0.05	0.19	-0.02	-0.06	0.09	1.00

(H) SNX									
	Binance	Bybit	OKX	Huobi	Gate	Bitget	dYdX	CoinEx	DeFi Loan
Binance	1.00	0.85	0.71	0.44	0.70	1.00	0.59	0.58	0.10
Bybit	0.85	1.00	0.74	0.49	0.64	0.85	0.57	0.63	0.11
OKX	0.71	0.74	1.00	0.41	0.52	0.71	0.51	0.65	0.07
Huobi	0.44	0.49	0.41	1.00	0.70	0.44	0.28	0.43	-0.02
Gate	0.70	0.64	0.52	0.70	1.00	0.70	0.49	0.46	0.02
Bitget	1.00	0.85	0.71	0.44	0.70	1.00	0.59	0.58	0.10
dYdX	0.59	0.57	0.51	0.28	0.49	0.59	1.00	0.40	0.11
CoinEx	0.58	0.63	0.65	0.43	0.46	0.58	0.40	1.00	0.11
DeFi Loan	0.10	0.11	0.07	-0.02	0.02	0.10	0.11	0.11	1.00

TABLE 2. Correlation of 8-hour funding rate time series from April to October 2023 between various platforms. The DeFi loan position's implied funding rates are calculated as Eq. 5 in a simulated environment with historical data on price and interest rates (see configurations in §5.2); funding rates of other platforms are directly fetched historical values. Red denotes positive correlation, blue negative; the stronger the shade, the higher the correlation.

5.4. Scenarios Analysis. Different from those orderbook-based platforms where funding rates are unrelated to a position's leverage ratio or direction (long / short), implied funding rates on lending platforms vary depending on these factors. In this section, we deviate from the basic setup as described in §5.2, and consider six scenarios: three leverage factors—2, 5, 20—and for both long and short loans.

Figure 8 shows that higher leverage leads to a bigger difference in funding rate between the long position and the short position.

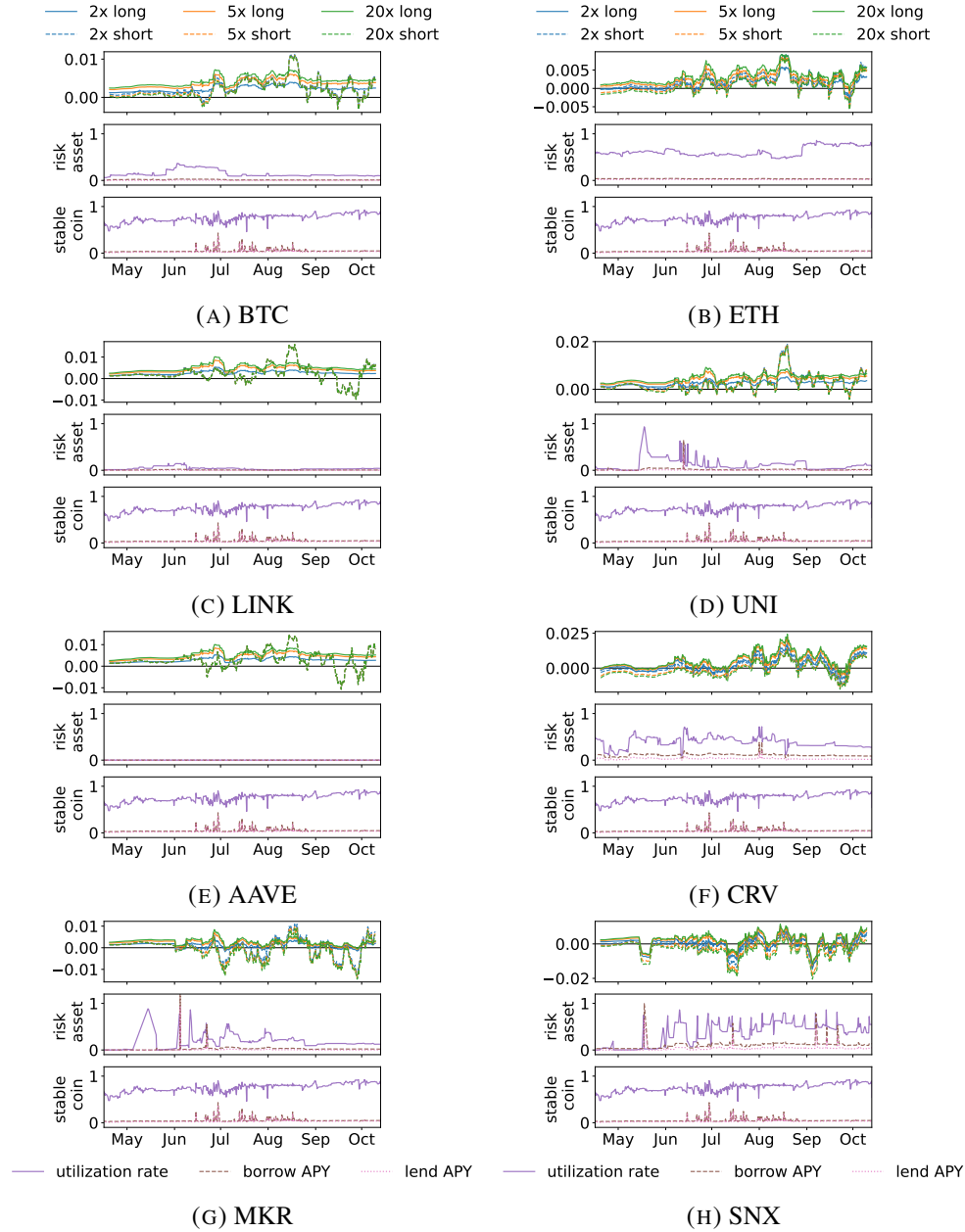


FIGURE 8. The DeFi loan position's implied funding rates, calculated as Eq. 5 in a simulated environment with historical data on price and interest rates from April to October 2023. For each asset, six scenarios are plotted: leverage factor of 2x (blue), 5x (orange) and 20x (green), and each when the risk asset is longed (solid line) and shorted (dashed line). Each line represents the moving 7-day average of 8-hour implied funding rates.

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